ZEPTO MYSQL DATA ANALYSIS PROJECT

PROJECT PROBLEM STATEMENT

1. Background

Quick-commerce platforms such as Zepto manage large product catalogs with changing prices, discounts, and inventory levels. Understanding this information is important for evaluating pricing strategies, product availability, and inventory performance. The project uses a product-level dataset containing 3,732 records and 9 columns to explore these business areas through SQL.

2. Problem Statement

The raw Zepto product dataset contains information about product categories, names, maximum retail prices, discounted selling prices, discount percentages, available quantities, product weights, and stock status. However, raw data may contain missing or invalid values and does not directly provide the business metrics needed to assess pricing and inventory performance. The problem is to clean, organize, and analyze this data using **MySQL and SQL** in order to identify useful patterns and generate actionable insights about product discounts, category-level revenue estimates, inventory availability, and product value.

3. Project Objectives

- Explore the dataset structure, record count, product categories, and stock availability.
- Identify missing, duplicate, and invalid product or pricing records.
- Clean price fields and convert values from paise to Indian rupees where applicable.
- Identify products with high discounts, high prices, and out-of-stock status.
- Estimate category-wise revenue using discounted selling price and available quantity.
- Compare product value using price-per-gram calculations.
- Classify products by weight and calculate category-wise inventory weight.
- Translate query results into clear business observations.

4. Scope of Analysis

The analysis covers database and table creation, data exploration, data cleaning, and eight SQL business analyses: best-value products by discount, high-MRP products that are out of stock, estimated revenue by category, high-priced products with low discounts, categories with the highest average discounts, price per gram, weight-based product classification, and category-wise inventory weight.

5. Tools and Technologies

Technology	Purpose
MySQL	Database creation, storage, and SQL execution.
SQL	Data exploration, cleaning, aggregation, and business analysis.
CSV	Source dataset containing product-level information.

6. Expected Outcome

The expected outcome is a cleaned and structured product dataset, supported by reusable SQL queries and business-focused results. The analysis should help explain how discounts vary across categories, which products are unavailable or offer better value, and how inventory is distributed across product categories. Revenue figures are estimates based on discounted selling price multiplied by available quantity; they should not be interpreted as

actual sales revenue.

7. Project Deliverable

A **MySQL SQL script** containing table creation, data exploration, cleaning operations, and the eight business analysis queries, together with the source CSV dataset.